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$$\begin{aligned} & \int \frac{\sqrt[4]{x} \sqrt[6]{x^7}}{\sqrt{x^3} \sqrt[12]{x^{11}}} dx \\ &= \int \frac{x^{\frac{1}{4} + \frac{7}{6}}}{\sqrt{x^{3 + \frac{11}{12}}}} dx \\ &= \int \frac{x^{\frac{17}{12}}}{x^{\frac{47}{24}}} dx \\ &= \int x^{\frac{34-47}{24}} dx = \int x^{\frac{-13}{24}} dx \\ &= \frac{x^{\frac{11}{24}}}{\frac{11}{24}} + C = \frac{24}{11} x^{\frac{11}{24}} + C \end{aligned}$$

References